«CUSTOMER NAME»
«GROUP COMPANIES»

Predictive Analytics · Service C

Predictive Analytics Run

Technical and Commercial Proposal

Monthly Monitoring, Retraining and Improvement of the Prediction Models of «CUSTOMER» in Operation

Proposal No: KOMTAS-«CUSTOMER ABBREVIATION»-PRA-C-«YEAR»-«SEQUENCE»

Proposal Date: «dd Month yyyy»

Validity: «proposal date + 1 month»

VARIABLE GLOSSARY — in this template only the fields inside « » change per customer. All other text is fixed.

Identity: «CUSTOMER» short customer name used in the text · «CUSTOMER NAME» full legal name (cover) · «GROUP COMPANIES» group companies in scope; delete the line for a single company · «CUSTOMER ABBREVIATION» short code in the proposal number · «YEAR» year · «SEQUENCE» proposal sequence number · «dd Month yyyy» proposal date · «proposal date + 1 month» end of validity · «price validity date» latest start date for which the fees stay valid

Context: «CUSTOMER CONTEXT» one paragraph: company or group, sector, customers or products concerned, where the data sits today, reason for the request

Use case: «FAMILY» use-case family: Retention & Churn, Demand & Supply, Pricing & Personalization, Risk & Collections or Industrial Operations · «USE CASE» use case from the catalogue in ANNEX-1 (e.g. asset churn prediction, demand forecasting, collection optimization) · «FAMILY — RETENTION & CHURN» block of the Retention & Churn family; keep it only if this family is in scope · «FAMILY — DEMAND & SUPPLY» block of the Demand & Supply family; keep it only if this family is in scope · «FAMILY — PRICING & PERSONALIZATION» block of the Pricing & Personalization family; keep it only if this family is in scope · «FAMILY — RISK & COLLECTIONS» block of the Risk & Collections family; keep it only if this family is in scope · «FAMILY — INDUSTRIAL OPERATIONS» block of the Industrial Operations family; keep it only if this family is in scope · «PLATFORM» delivery platform agreed with the customer: Google Cloud, Dataiku, Microsoft Azure or an open-source stack · «FREQUENCY» how often predictions or retraining are produced (e.g. monthly, weekly, daily) · «HORIZON» forecast horizon (e.g. 4 weeks, 13 weeks)

Scope: «PILOT REF» proposal or project number of the Pilot (A) whose model was validated; delete the reference if there is none · «PRODUCTION REF» proposal or project number of the Production (B) that put the models into operation; delete the reference if the models were not deployed by Komtaş · «SUPPORT HOURS» support hours (e.g. weekdays 09:00–18:00 Türkiye time) · «BAND» sizing band S, M or L from the proposal input form

Commercial: «fee» fee excl. VAT · «monthly fee» monthly fee excl. VAT · «amount» payment amount excl. VAT · «day rate» person-day rate (change requests) · «hour rate» hourly rate for work above the monthly capacity · «CITY» city where on-site meetings are free of charge

Values: «NUMBER» numeric value (count, threshold, days, weeks, months, hours)

Annex: «RESULT» measured result of a reference project, only if measured and approved for use; otherwise delete the line

Contacts: «full name» contact name · «title» contact title · «e-mail / phone» contact details

Markers: of the paragraphs, rows and annex pages starting with «FAMILY — RETENTION & CHURN», «FAMILY — DEMAND & SUPPLY», «FAMILY — PRICING & PERSONALIZATION», «FAMILY — RISK & COLLECTIONS» and «FAMILY — INDUSTRIAL OPERATIONS», keep those of the families in operation and delete the others; then delete the family markers. The orange AX DOĞRULASIN and GÜNER DOĞRULASIN tags mark draft statements to be verified before use. FILL NOTE boxes are internal and are deleted before sending.

Contents

FILL NOTE — this template is for service C (Run) of the Predictive Analytics package: the monthly monitoring, retraining and improvement of the models in operation. It is normally used after service B. It can also be used for an equivalent model put into operation by «CUSTOMER» or by another party; in that case keep the second paragraph of Section 4.1 and delete the reference to «PRODUCTION REF». «AX DOĞRULASIN»

Of the paragraphs, rows and annex pages starting with «FAMILY — RETENTION & CHURN», «FAMILY — DEMAND & SUPPLY», «FAMILY — PRICING & PERSONALIZATION», «FAMILY — RISK & COLLECTIONS» and «FAMILY — INDUSTRIAL OPERATIONS», keep those of the families in operation (one or more) and delete the others; then delete the remaining family markers, replace «FAMILY» and «USE CASE» with the families and use cases in operation and «PLATFORM» with the platform they run on.

Statements followed by an orange tag are draft assumptions of the package. AX DOĞRULASIN: verified with the Analytics team (capacity, activities, frequencies, thresholds, metrics, priorities, roles, term, references). GÜNER DOĞRULASIN: verified with Cloud Solutions (cloud services, cloud architecture, cloud access, consumption). After verification only the tag is deleted. No proposal is sent with a tag in it.

Order of completion: models in scope, platform and band from the proposal input form (Table 2) → drift thresholds (Table 5) → retraining schedule (Table 6) → support hours and response times (Table 9) → effort and fees (Section 9) → term (Section 11) → remaining placeholder fields. Margin, discount or incentive figures are never written into this proposal.

1. Purpose

«CUSTOMER CONTEXT»

The aim of this proposal is to keep the prediction models of «CUSTOMER» in operation accurate and running, month by month, from a fixed monthly capacity of hours: their data and performance are monitored, they are retrained on schedule or when their performance drops, and they are improved within the capacity. The models were put into operation in the Production project «PRODUCTION REF», on the basis of the Pilot «PILOT REF».

The models run on «PLATFORM». On every platform the same functions are used: data preparation, model training, a model registry with versions, scheduled pipelines that write the scores and forecasts, export of the results to the business systems and monitoring reports.

On Google Cloud the data is in BigQuery, the models are trained with BigQuery ML or Vertex AI and kept in the Vertex AI model registry, scheduled pipelines write the results to BigQuery tables and the monitoring reports are in Looker. On Microsoft Azure the same functions are provided by Azure Machine Learning and Azure Data Factory or Microsoft Fabric. «GÜNER DOĞRULASIN»

On Dataiku the models are kept and retrained in Dataiku projects, with scenarios for scheduled and conditional retraining and for the monitoring of data and model stability; on an open-source stack the same functions are built with open-source tools. «AX DOĞRULASIN»

The package is offered as three services. This proposal covers service C.

Service	Focus	Result	Pricing
A — Pilot	One use case: data readiness, feature set, model, validation, value measurement	Validated model and go / no-go decision	Fixed fee (separate proposal)
B — Production	Deployment pipeline, scheduled scoring, model registry, monitoring, integration	Model in operation, results in business systems	Fee per use case (separate proposal)
C — Run	Monitoring, retraining and improvement of the models in operation	Models kept accurate over time	Monthly fee

Table 1 — Service structure.

2. Scope

2.1 Models in Scope

Item	Scope
Families and use cases in operation	«FAMILY» — «USE CASE»
Platform	«PLATFORM»
Number of models in operation	«NUMBER»
Data sources feeding the models	«NUMBER»
Volume: customers, products, time series, stores or process variables, depending on the use case	«NUMBER»
Prediction frequency	«FREQUENCY»
Business systems receiving the results	«NUMBER»
Support hours	«SUPPORT HOURS»
Monthly capacity	«NUMBER» hours
Sizing band	«BAND»

Table 2 — Models in scope.

If the number of models, data sources or business systems changes by more than «NUMBER», the band and the monthly capacity are reassessed.

2.2 Service Components

The service has the components below. «AX DOĞRULASIN»

Area	Content
Run monitoring	Follow-up of the scheduled scoring and forecast runs and of the exports to the business systems; failed runs restarted and their cause fixed
Drift monitoring	Distribution of the input data and of the predictions compared with the training period, against the thresholds in Table 5
Performance monitoring	Technical metrics of each model computed on the latest observed outcomes and compared with the validation baseline
Retraining	Scheduled retraining (Table 6) and retraining when a threshold is crossed; the new version is promoted only after comparison with the current version
Monthly report	Runs, drift, performance, retraining, incidents, use of the capacity and recommendations
Small improvements	Changes requested by «CUSTOMER» that fit within the monthly capacity (e.g. a new feature, a changed threshold, a new report field, a new export column)

Table 3 — Service components.

2.3 What Is Monitored per Family

The metrics monitored for each family are: «AX DOĞRULASIN»

Family	Technical Metrics	Business Metric (reported)
«FAMILY — RETENTION & CHURN» Retention & Churn	AUC or Gini and lift in the top decile on the customers whose outcome is known	Churn or segment downgrade of targeted customers against a control group, if campaigns keep a control group
«FAMILY — DEMAND & SUPPLY» Demand & Supply	WAPE / MAPE and bias per horizon on the latest periods	Stock-outs, overstock or stock holding cost, if the data is shared
«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization	precision@k and recall@k on the latest interactions; fit of the price elasticity on the latest periods	Revenue, conversion or click-through rate of a test group, if tracked
«FAMILY — RISK & COLLECTIONS» Risk & Collections	Gini and stability of the score distribution on the latest observed outcomes	Transition into delinquency or collection result against a control group, if kept
«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations	Prediction error of the process variable or accuracy of the classification on the latest periods	Energy consumption, deviation, quality or yield, if the data is shared

Table 4 — Metrics per family. Delete the rows of the families not in operation.

2.4 Boundaries

- New use cases and new models are built in services A and B; Run covers the models in Table 2. «AX DOĞRULASIN»
- The source systems and the data platform remain with «CUSTOMER»; Komtaş follows the data that feeds the models and reports issues to the data owner. «AX DOĞRULASIN»
- Actions based on the results (e.g. campaigns, orders, prices, collection contacts, process settings) are taken by «CUSTOMER».
- The analytics organisation and the model platform of «CUSTOMER» are the subject of the Analytic CoE Enablement Program and of MLOps platform work.
- The governance of AI and models across the organisation is the subject of AI Governance Readiness.

3. Approach

3.1 Operating Model

Changes to the pipelines, features and models are made in the code repository or in the project of the platform and deployed through the deployment pipeline set up in service B. Each model version is kept in the model registry with its training data period and its metrics. Requests and incidents are recorded in a ticket system agreed in the onboarding period; each record has a priority, an owner and a status. «AX DOĞRULASIN»

3.2 Drift and Performance Thresholds

The thresholds below are agreed in the onboarding period and recorded in the runbook: «AX DOĞRULASIN»

Signal	Threshold	Action
Drift of the input data (distribution of key features)	«NUMBER»	Alert, analysis, note in the monthly report
Drift of the predictions (distribution of scores or forecasts)	«NUMBER»	Alert, analysis, note in the monthly report
Drop of the technical metric against the validation baseline	«NUMBER»	Retraining outside the schedule
Missing or late input data	«NUMBER» hours	Incident; data owner informed
Failed scoring or forecast run	—	Incident; run restarted after the cause is fixed

Table 5 — Drift and performance thresholds.

3.3 Retraining Schedule

Family	Scheduled Retraining	Additional Trigger
«FAMILY — RETENTION & CHURN» Retention & Churn	«FREQUENCY»	Threshold in Table 5 crossed; new products or segments
«FAMILY — DEMAND & SUPPLY» Demand & Supply	«FREQUENCY»	Threshold in Table 5 crossed; new stores, sites or products
«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization	«FREQUENCY»	Threshold in Table 5 crossed; major catalogue or price list change
«FAMILY — RISK & COLLECTIONS» Risk & Collections	«FREQUENCY»	Threshold in Table 5 crossed; change of credit or collection policy
«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations	«FREQUENCY»	Threshold in Table 5 crossed; change of equipment, recipe or operating mode

Table 6 — Retraining schedule. Delete the rows of the families not in operation.

> *FILL NOTE — draft retraining frequencies:*
> *Retention & Churn monthly or quarterly · Demand & Supply monthly · Pricing & Personalization weekly for recommendation, monthly for the other use cases · Risk & Collections and Industrial Operations agreed in the onboarding period.* «AX DOĞRULASIN»

A retrained model replaces the current version only if its metrics on the same validation period are at least equal; the result is recorded and shown in the monthly report. «CUSTOMER» can ask to approve each replacement. «AX DOĞRULASIN»

3.4 Division of Responsibilities

Responsibilities are divided as follows: «AX DOĞRULASIN»

Area	Komtaş	«CUSTOMER»
Scoring and forecast runs	Follow-up, restart, fixes in the pipelines	—
Input data	Checks and alerts	Owner; delivery on schedule; fixes in the source systems
Models	Monitoring, retraining, versions, small improvements	Approval of thresholds and, where requested, of new versions
Outcome data (e.g. actual churn, sales, repayments, process readings)	Use in performance monitoring	Owner; sharing on schedule
Business actions based on the results	—	Owner
Cloud account, platform licences and billing	Notes on consumption in the monthly report	Owner; payment to the provider

Table 7 — Division of responsibilities.

3.5 Monthly Capacity

The monthly capacity is «NUMBER» hours. It covers run monitoring, drift and performance monitoring, retraining, the monthly report, incident work and small improvements. Unused hours are not carried over to the next month. «AX DOĞRULASIN»

Work above the monthly capacity is carried out only after written approval of «CUSTOMER», at the hourly rate in Section 9. An improvement that needs more than «NUMBER» hours is estimated and proposed separately. «AX DOĞRULASIN»

4. Activities and Timeline

4.1 Start of Service

The service starts with an onboarding period of «NUMBER» weeks: access, runbooks, thresholds, alert routing, the ticket system and the contact and escalation list are agreed, and the baseline metrics of each model are recorded. «AX DOĞRULASIN»

If the models were not put into operation by Komtaş, the onboarding period also includes a review of the models, the pipelines and the documentation. Findings that prevent operation are fixed before the start of service, as a separate item. «AX DOĞRULASIN»

4.2 Monthly Cycle

The monthly cycle is: «AX DOĞRULASIN»

Activity	Content	Frequency
Run monitoring	Scoring and forecast runs, exports to business systems	Each run, within the support hours
Drift monitoring	Input data and prediction distributions against Table 5	Each run or weekly
Performance monitoring	Technical metrics on the latest observed outcomes	Monthly
Retraining	Training, comparison with the current version, promotion	Table 6
Monthly report and review meeting	Runs, drift, performance, retraining, incidents, capacity, requests	Monthly
Small improvements	Request → estimate → approval by «CUSTOMER» → change through the repository → check	On request, within the capacity

Table 8 — Monthly cycle.

4.3 Support Hours and Priorities

Support is given within «SUPPORT HOURS». Incidents are handled by priority:

Priority	Definition	Response Time
P1 — Critical	Results not delivered to a business system that uses them daily; no workaround «AX DOĞRULASIN»	«NUMBER» hours
P2 — High	Run failed or delayed with a workaround (e.g. previous scores still usable); threshold crossed on a production model	«NUMBER» hours
P3 — Low	Question or request without effect on operation	«NUMBER» working days

Table 9 — Priorities and response times.

Response time is measured from the record of the incident in the ticket system to the first response of the Komtaş engineer.

5. Deliverables and Acceptance

5.1 Deliverables

Documents are delivered in Turkish, in editable format (Microsoft Word / Excel / PowerPoint) together with a PDF copy.

#	Deliverable	Frequency	Content
D1	Onboarding report	Once	Access, runbooks, thresholds, alert routing, baseline metrics; findings of the model review if any
D2	Runbook	Kept up to date	Operating procedures: restart of runs, retraining, promotion of a version, incident handling
D3	Monthly report	Monthly	Runs, drift, technical metrics per model, retraining and versions, incidents, use of the capacity, recommendations «AX DOĞRULASIN»
D4	Retraining record	Per retraining	Training data period, metrics of the new and current version, decision
D5	Incident report	Per P1 incident	Timeline, cause, resolution and measures against recurrence

Table 10 — Deliverables.

5.2 Acceptance

The monthly report is reviewed by «CUSTOMER» within «NUMBER» working days. If no written objection is received in that period, the report and the hours used are deemed accepted.

The response times in Table 9 are the service levels of this proposal. The technical metrics are reported against the thresholds in Table 5; the accuracy of the models and the business result depend on the data and on the actions of «CUSTOMER» and are not guaranteed. «AX DOĞRULASIN»

6. Customer Responsibilities

«CUSTOMER» is responsible for the items below. A delay in any of them can affect the service levels.

- Appoint a service owner who approves requests, thresholds and work above the capacity.
- Deliver the input data on the agreed schedule and inform Komtaş in advance of changes in the source systems.
- Share the outcome data needed for performance monitoring (e.g. actual churn, actual sales, repayments, clicks and purchases, process readings) on the agreed schedule. «AX DOĞRULASIN»
- Provide the access needed for operation: roles on the cloud projects or subscriptions of the models, access to the platform projects and write access to the code repository. «GÜNER DOĞRULASIN»
- Keep the legal basis under KVKK for using customer data in the models. «AX DOĞRULASIN»
- Keep the cloud account, the platform licences and the contract with the cloud provider active.

7. Assumptions and Out of Scope

7.1 Assumptions

The capacity and fee in this proposal are based on the assumptions below. If they do not hold, the impact is assessed through a change request.

- The models in scope match Table 2.
- The models, pipelines and features are kept in a code repository or platform project and deployed through a pipeline; parts built outside it are brought into it in the onboarding period or are excluded from the service. «AX DOĞRULASIN»
- Scoring and forecasting are batch runs; real-time (online) scoring is not part of the service unless stated in Section 2. «AX DOĞRULASIN»
- The work is carried out remotely; on-site work in «CITY» is agreed case by case. «AX DOĞRULASIN»
- The working language and the language of the deliverables is Turkish.

7.2 Out of Scope

The topics below are outside the scope of this proposal. If requested, they are assessed and proposed separately. «AX DOĞRULASIN»

- New use cases and new models (services A and B).
- Migration of the data platform or of the analytics platform.
- Cleansing of data in the source systems.
- Execution of campaigns, orders, price changes, collection actions or process changes based on the results.
- A guaranteed accuracy of the models or business result.
- Support outside «SUPPORT HOURS».
- Cloud consumption and licence costs.

8. Team

Role	Responsibility	Involvement
Run Lead	Monthly report and review meeting, request management, decisions on retraining and versions, escalation	Monthly, part time «AX DOĞRULASIN»
ML Engineer	Run and drift monitoring, retraining, pipelines, incidents, small improvements	Within the monthly capacity

Table 11 — Komtaş team and roles.

- A second engineer who knows the models is named as backup. «AX DOĞRULASIN»

- If a change in the team is needed, the replacement has at least equivalent qualifications and is notified to «CUSTOMER» in advance.

9. Effort and Fees

> *FILL NOTE — take the band and the monthly capacity from the proposal input form (predictive-analytics-3c-run-girdi-formu). Fees are entered by sales. Draft values:*
>
> *Monthly capacity: S 40 hours · M 80 hours · L 160 hours. Band drivers: use cases in operation S 1 · M 2 · L 3; data sources S ≤3 · M 4–8 · L >8; volume S ≤100k customers or ≤10k series · M up to 1M or 100k · L above; prediction frequency S monthly · M weekly · L daily; systems integrated S 1 · M 2–3 · L >3. Band rule: the highest band reached by at least three of the five drivers. Minimum term: 12 months.* «AX DOĞRULASIN»
>
> *The values are not based on recorded effort and are updated with the data of the Analytics team. Delete this box before sending.*

Run is charged as a monthly fee for the monthly capacity in Table 2.

Item	Band	Capacity	Fee (excl. VAT)
Run	«BAND»	«NUMBER» hours per month	«monthly fee» per month
Onboarding period	—	«NUMBER» person-days	«fee»
Work above the monthly capacity	—	Per hour	«hour rate»
Change requests	—	Per person-day	«day rate»

Table 12 — Fees. All fees exclude VAT.

Payment Item	Amount (excl. VAT)
Onboarding period — at its end	«amount»
Run — monthly, at the end of each month	«monthly fee»
Work above the capacity — monthly, with the hours approved	«amount»

Table 13 — Payment schedule. All fees exclude VAT.

Cloud consumption and licence costs are not part of this proposal.

10. Next Step

Run can be extended as the use of the models grows. The usual follow-on work is:

Need	Follow-On Work
Other business questions	Pilot and production of a further use case from the same or another family (ANNEX-1)
More models, sources or business systems in operation	Higher band and monthly capacity of Run
Data in on-premises ETL tools or warehouses	ETL Modernization on Google Cloud, Data Modernization
Analytics organisation and model platform	Analytic CoE Enablement Program, MLOps platform
Governance of the models	AI Governance Readiness
Cloud consumption	Purchase of Google Cloud or Microsoft Azure consumption through Komtaş

Table 14 — Follow-on work.

10.1 Further Use Cases

A further use case starts with a pilot (service A) and is put into operation in service B. Once it is in operation, it is added to Run and the band is reassessed.

10.2 Data Platform

Where part of the data that feeds the models sits in on-premises ETL tools or warehouses, its migration to a cloud data platform (e.g. Google Cloud) shortens the data delivery and simplifies the pipelines. «GÜNER DOĞRULASIN»

10.3 Consumption

Komtaş is a Google Cloud Premier partner and has completed its partnership with Microsoft Azure. The cloud consumption of «CUSTOMER» can be purchased through Komtaş; the commercial terms are set out in a separate proposal. «GÜNER DOĞRULASIN»

11. Commercial Terms

11.1 General Terms

- All fees in this proposal are stated in Turkish lira and exclude VAT. Statutory taxes, duties and charges are not included in the fees.
- The proposal is valid until «proposal date + 1 month».
- The fees are valid on condition that the service starts by «price validity date». If the start moves beyond that date for any reason, the fees and pricing terms are reassessed by mutual agreement in the light of the economic and financial conditions at the time of the work.
- Run has a minimum term of «NUMBER» months, starting at the end of the onboarding period. After the minimum term it continues month by month and can be ended by either party with «NUMBER» months' written notice.
- The monthly fees are reviewed once a year by mutual agreement.
- Payment follows the payment schedule in Section 9. Invoices are payable within «NUMBER» days of the invoice date.
- Scope changes are managed through a written change request; no change is implemented before its impact analysis is approved.
- The parties keep confidential all information they access within the scope of the work. A separate non-disclosure agreement is signed where needed.
- At the end of the service, the runbook, the code repository, the model versions and the access are handed over to «CUSTOMER» or to the party it names.
- This proposal forms an integral annex to the contract to be signed by the parties. In the event of a conflict between the contract and the proposal, the contract prevails.

11.2 Proposal Contact Details

Questions on this proposal and its scope can be addressed to the contacts below.

Area of Responsibility	Full Name	Title	Contact
Commercial	«full name»	«title»	«e-mail / phone»
Technical	«full name»	«title»	«e-mail / phone»

11.3 Proposal Annexes

This proposal is submitted with the following annexes:

- ANNEX-1 — Use-Case Family Pages
- ANNEX-2 — About Komtaş

ANNEX-1 — Use-Case Family Pages

> *FILL NOTE — keep only the pages of the families in operation (Table 2) and delete the others.*

The use cases of Predictive Analytics are grouped into five families. Every use case follows the same lifecycle: data readiness check → feature set → model → validation → deployment → monitoring. The lifecycle does not change with the delivery platform. The page of each family in scope is given below; field references and measured results are listed in Table 21 (ANNEX-2).

«FAMILY — RETENTION & CHURN» Retention & Churn

The use cases, data, outputs and success metrics of the Retention & Churn family are: «AX DOĞRULASIN»

Item	Content
Use cases	Path to Churn (action on the critical steps before the decision to churn); "balance" churners (incoming funds withdrawn after a few days); asset churn and segment downgrade (banking); subscription and membership churn; customer satisfaction index.
Problem	Customers leave, cancel or move their assets; retention actions reach them late or are spread over all customers at the same cost.
Typical sectors	Banking, fitness and subscription services, telecommunications, retail loyalty programmes, insurance.
Data	At least 12–24 months of customer, product or contract, transaction, visit or usage, campaign and complaint data; in banking, the history of assets and segments.
Label	Churn, cancellation or segment downgrade as defined with the business owner (e.g. no renewal, or no visit for «NUMBER» days).
Output	Probability per customer and the main drivers behind it, refreshed «FREQUENCY»; critical events passed to sales or marketing as alert-action scenarios.
Success metrics	AUC or Gini and lift in the top decile on hold-out data. Business metric: difference in churn or segment downgrade between the targeted customers and a control group.

Table 15 — Retention & Churn family.

«FAMILY — DEMAND & SUPPLY» Demand & Supply

The use cases, data, outputs and success metrics of the Demand & Supply family are: «AX DOĞRULASIN»

Item	Content
Use cases	Demand forecasting (retail, manufacturing); inventory productivity analysis; visitor and sales prediction (shopping malls); schedule optimization.
Problem	Stock-outs, overstock and planning errors arise when the quantities ordered, produced or allocated do not follow demand.
Typical sectors	Retail, e-commerce, FMCG, manufacturing, distribution, shopping mall management.
Data	At least 2–3 years of sales or demand at product–store or product–site and day or week level, prices and promotions, stock levels and stock-outs, calendar and holidays; external data where relevant (e.g. weather, price indices, visitor counts).
Output	Forecast per series and horizon («HORIZON»), refreshed «FREQUENCY»; scenario runs where the business needs them.
Success metrics	WAPE / MAPE and bias against the current method of «CUSTOMER» on a back-test period. Business metric: change in stock-outs, overstock or stock holding cost.

Table 16 — Demand & Supply family.

«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization

The use cases, data, outputs and success metrics of the Pricing & Personalization family are: «AX DOĞRULASIN»

Item	Content
Use cases	Next best offer; product bundling; recommendation; customer lifetime value; deposit pricing; credit pricing; list price and markdown optimization; dynamic pricing; customer segmentation; customer contact policy; onboarding analytics.
Problem	Offers, prices and contacts are the same for all customers or follow fixed rules; they do not follow the response of each customer or segment.
Typical sectors	Banking, retail, e-commerce, spare parts and distribution, media.
Data	At least 6–12 months of transactions, offers and responses, prices, discounts or negotiation records, the product catalogue and hierarchy and customer attributes; for recommendation, interactions (views, baskets, purchases).
Output	Offer, product list, price or segment per customer or per product group and customer segment, refreshed «FREQUENCY»; for pricing, the price elasticity per product group and customer segment.
Success metrics	precision@k and recall@k offline; click-through rate, conversion or revenue in an A/B test against the current offer or price. Business metric: revenue or conversion of a test group against a control group.

Table 17 — Pricing & Personalization family.

«FAMILY — RISK & COLLECTIONS» Risk & Collections

The use cases, data, outputs and success metrics of the Risk & Collections family are: «AX DOĞRULASIN»

Item	Content
Use cases	Collection analytics and collection optimization; credit risk model enhancement, including graph analytics on relational data; dynamic client limit; risk analytics.
Problem	Collection contacts and credit decisions apply the same rules to customers with different levels of risk; existing models do not use the relations between customers.
Typical sectors	Banking and other lenders.
Data	Credit, repayment and delinquency history, limits and their use, customer and product data, contact history of the collection channels (call centre, IVR, SMS / push); for graph analytics, payments, transfers and links between customers (e.g. partnership, guarantorship).
Label	Delinquency within the next «NUMBER» days, or a default definition agreed with risk management.
Output	Probability of delinquency or default per customer, with the importance and direction of effect of the main features; contact policy or limit distribution per risk level.
Success metrics	Gini on hold-out and out-of-time data. Business metric: transition into the first delinquency cycle against a control group, or credit approval rate at the same NPL ratio.

Table 18 — Risk & Collections family.

«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations

The use cases, data, outputs and success metrics of the Industrial Operations family are: «AX DOĞRULASIN»

Item	Content
Use cases	Energy optimization (HVAC systems, furnaces); predictive quality and quality classification; production yield optimization; predictive maintenance; process unit prediction; fixed asset utilization.
Problem	Processes are run with settings chosen by experience; energy use, deviations, defects and downtime are seen only after they occur.
Typical sectors	Manufacturing (e.g. glass, pharmaceuticals), refining and petrochemicals, other energy-intensive production.
Data	Sensor and IoT time series of the process (temperatures, flows, energy, set values), laboratory samples, quality, maintenance and production records; controllable and uncontrollable factors identified with the process engineers.
Output	Prediction of the process variable (e.g. energy, temperature, quality class, yield) and recommended settings within the operating constraints, based on ML-guided simulation.
Success metrics	Prediction error of the process variable or accuracy of the classification on hold-out periods. Business metric: energy consumption, deviation, quality or yield of test units or periods against control units or periods.

Table 19 — Industrial Operations family.

ANNEX-2 — About Komtaş

> *FILL NOTE — the figures in Table 20 and in the bullets below come from the proposals of 2026. Check them against the current corporate profile before sending.*

Komtaş is an independent information technology company that has specialised in enterprise data management in Türkiye since 1989. More than 35 years of focus on a single field has given it an end-to-end view that follows data from the moment it is created to the moment a decision is taken.

Indicator	Value
Founded	1989 — more than 35 years of continuous data-focused activity
Specialist staff	250+ data engineers, architects, consultants and analysts
Offices	Istanbul, Ankara, Izmir, Dubai
Strategic partnerships	10+ global technology vendors
Focus area	Data governance, data architecture, analytics, artificial intelligence and AI agent based solutions
Positioning	"From Data to Agents" — integrated capability from data to AI agents

Table 20 — Komtaş corporate profile.

Service Capabilities

The service portfolio of Komtaş covers the whole life cycle of a data initiative:

- **Data Governance and Data Management:** data governance maturity assessment, framework and organisation design, policy and process design, data catalogue, business glossary, data quality management, master data management, data privacy and classification.
- **Data Architecture and Engineering:** data warehouse and data lake architectures, lakehouse design, data integration and streaming architectures, cloud and on-premises modernisation.
- **Analytics and Artificial Intelligence:** enterprise reporting, advanced analytics, machine learning models, generative AI applications and AI governance.
- **Managed Services and Support:** P1 / P2 / P3 support for production systems, capacity and performance management, continuous improvement services.
- **Training and Capability Building:** product training, role-based data governance training, train-the-trainer programmes.

Platforms and Partnerships

Komtaş is a Google Cloud Premier partner. Its partnership with Microsoft Azure has been completed.

The use cases of this package are delivered on the platform agreed with the customer: Google Cloud, Dataiku, Microsoft Azure or an open-source stack. The lifecycle and the services are the same on every platform.

Predictive Analytics References

Reference projects of Komtaş in the five families of this package, with the results measured in them, are listed below. Reference calls with the customers concerned can be arranged on request. «AX DOĞRULASIN»

FILL NOTE — keep the rows of the family in scope or all rows. Do not change the results and do not add customer names, contact persons or results that are not approved for use in proposals.

Family	Use Case	Organisation	Result
Retention & Churn	Asset churn prediction (segment downgrade)	Odeabank	11% lower segment downgrade in the test group than in the control group
Demand & Supply	Demand prediction	A tyre manufacturer	7% lower stock holding cost for 7 tyre types
Demand & Supply	Visitor and sales prediction	A shopping mall management company	Forecasts for 350+ stores in 2 malls with 1.6 million visitors
Pricing & Personalization	List price and markdown optimization	A spare-parts seller	+4% revenue at the same conversion level in the test group; price elasticity for 30,000+ SKUs; 12 A/B test scenarios
Risk & Collections	Collection optimization	Albaraka Banking Group, TEB	3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; optimized contact policy
Risk & Collections	Credit risk model enhancement with graph analytics	QNB Finansbank	+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81
Industrial Operations	Energy optimization in HVAC systems	Eczacıbaşı Monrol	2.3% lower energy consumption in the test units than in the control group
Industrial Operations	Furnace energy optimization, furnace temperature control, glass quality classification, production yield optimization	Şişecam	«RESULT»
Industrial Operations	Continuous catalytic reformer (CCR) unit prediction	An integrated oil and petrochemical group	Qualitative: less error-prone operation of the unit and more control over the process

Table 21 — Predictive analytics references of Komtaş.

Komtaş has also built a subscription churn model for a fitness / subscription company (Retention & Churn) and a demand forecasting model for a retail company (Demand & Supply). Result: «RESULT». «GÜNER DOĞRULASIN»

Why Komtaş

- **Focus:** Komtaş has worked only in the data field for more than 35 years, with more than 250 specialists and 4 offices.
- **One method for five families:** retention, demand, pricing, risk and industrial use cases follow the same steps, from the data readiness check to monitoring.
- **Measured results:** the references in Table 21 give the results measured in the work, where they are available.
- **Data and model together:** the same organisation prepares the data, builds the models and operates them, on the platform of the customer.
- **From pilot to operation:** pilot, production, monthly operation and cloud consumption are provided by the same organisation (Section 10).